Digital Sovereignty and Resilience

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***Abstract*—Resilience in communication networks has been a field of study for decades. However, resilience has been studied by assuming that network actors are benign. Increasingly, digital sovereignty, where nation states and territories exercise jurisdiction over data residing or flowing in geographical areas in their control, is playing a role in the thinking of network design. In this paper, we scan some of the issues surrounding digital sovereignty and examine its interaction with traditional approaches to resilience. We describe some open problems in the area, such as developing a sovereignty metric that can interact with availability metrics in communication networks.**

***Index Terms*—Digital Sovereignty, Resilience**

I. Introduction

Governments are becoming increasingly concerned with sovereignty in the digital/cyber domain [1], [2], as many basic functions of government (e.g., tax collection) are now digital. The major issues revolve around how to ensure the security, privacy and availability of government and citizens digital information in any form, from sensing/creation of information to communication through networks to processing on computing platforms to storage in the cloud. Most nations do not have the wherewithal to domestically build all the hardware and software components that comprise the critical digital infrastructure upon which their digital services depend. Furthermore, this digital infrastructure upon which society depends is typically constructed and operated by private companies. Hence policy and regulatory issues to safeguard sovereignty have come to the forefront. For example, several countries (e.g., US, UK) have banned 5G wireless cellular equipment from Chinese companies over security/spying concerns [3].

The European Union has proposed addressing digital sovereignty in part by a mix of technical and regulatory efforts. On the technical side, the focus has been on ensuring data confidentiality and integrity through standards conforming verifiable hardware, adoption of open source software and localized data processing/storage with the European Federated Cloud Initiative [4]. Whereas on the policy side, General Data Protection Regulation (GDPR) and associated rules have put forth a legal framework to establish liability and enforce data protection.

One aspect of digital sovereignty that has not been adequately studied is resilience to failures, supply chain issues and hardware/software limitations. Here we explore how resilience techniques such as diversity and redundancy interact with the sovereignty aspects of digital infrastructure with the goal of informing policy and regulatory bodies. For example, adopting diversity techniques such as moving from a single hardware vendor data center to a multi-vendor center will improve the cyber resilience, but does it improve the sovereignty? What are the cost/benefit trade-offs? Such an approach may enable formerly banned vendors to be incorporated to increase diversity and allow them to be monitored for suspicious/malicious activity. Similarly, redundancy can be leveraged in certain situations to improve sovereignty. For example, there are four main vendors that build and maintain undersea cables. When using cables constructed or operated by an untrustworthy organization, one could establish redundant connections either via more trustworthy alternate cables or trustworthy alternate technology (e.g. satellites) to provide a backup communication path and possibly check for malicious activity.

The remainder of the paper is organized as follows. Section II discusses required background and relevant literature. Section III discusses illustrative examples of the interaction among sovereignty and resilience issues. Section IV discusses open problems and some possible lines of research. Lastly in Section V, we summarize our work and conclusions.

II. Background and Literature Review

In this section, we cover several important questions related to sovereignty. We will start with, what is digital sovereignty, try to understand how it is different or correlated with sovereignty, what are some example incidents that contributes to it, is digital sovereignty good or bad for us, how can we measure the resilience aspects of digital sovereignty, how to enforce digital sovereignty for a particular country? etc.

A. Digital Sovereignty

The online Merriam Webster dictionary[1] defines *sovereignty* as "supreme power over a body politic", where body politic refers to "a group of persons politically organized under a single governmental authority". A second meaning, which is better suited to our purpose is "freedom from external control". In other words, a nation or region would like its digital ecosystem to be free from being adversely or otherwise affected by forces or factors that are external to it. Conversely, it can be thought of as the ability to *exert control* over those digital services that are deemed essential or critical for the security and interests of the state and it's citizens.

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[1]https://m-w.com

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The term "Digital Sovereignty" has seen increased use in recent years because of where data pertaining to humans and national interests now reside and how such data flows across networks. Thus, actual control of data and data flow is rarely exclusive to the entity to whom such data might belong. Several factors impact such control or freedom from external control. We discuss some of them below.

- **Physical Dependencies and Alliances**: The influence of historical bonds, alliances and geography have played a significant role in the adoption of standards and even a way of life. While a country is free (in theory) to pick its own networking protocols, components and infrastructure, it is often impossible to ignore the underlying dependencies, history, geography and alliances. People in many of the former colonies of UK drive on the left side of the road unlike most parts of the world. The USA and Europe have differed in their implementation of the digital transmission hierarchy (T1 Vs E1) and even electricity (60 Hz AC Vs 50 Hz). In the second generation of wireless standards (2G), most of Europe agreed upon the Global System of Mobile Communications (GSM) whereas the USA, Canada and Mexico in North America started with the IS-54/136 and IS-95 standards. These were, in part geographically driven arrangements, although the economics of GSM finally brought it to North America. One could argue that Canada and Mexico, through their adjacency to the larger USA market and the historic connections with the connected public switched telephone network, have less sovereignty in their choice of technology.
- **Global Standards**: In many technologies (e.g., cellular wireless, such as 5G), international standards bodies (standard setting organizations - SSOs) are comprised of members from nation states [5]. However, the participants are from companies from these different nations. The influence of companies on standards process and nation states on companies affects digital sovereignty as so-called standard essential patents (SEPs) may belong to companies subject to the influence of a nation state [6]. Thus, it is important for a nation concerned about its digital sovereignty to aggressively participate in SSOs and have contributions to SEPs from its own companies. However, as previously mentioned, most nations are unlikely to have the wherewithal to be major players, and tend to adopt widely used (standard) technologies. In this sense, most nations yield their sovereignty to major players in standardization.
- **Vendor Diversity and Cyber Dependencies:** As previously mentioned, there are only four major vendors that build undersea cables for communication. Similarly, there are primarily three companies that build the core components of 5G cellular networks. In the latter case, the components rarely inter-operate smoothly making it necessary for service providers to adopt a single vendor for their infrastructure. The consequence of this lack of diversity is obvious, in that nations and their people are beholden to a small number of vendors for the critical digital communication infrastructure. If vendors are not transparent or accountable, or are controlled by their home nations, the digital sovereignty of a client nation is at a severe risk. Especially, in the case of 5G networks [6], equipment is physically located in a sovereign nation but manufactured elsewhere, and technical support, software updates, etc. need involvement of vendors from China or Europe.

  In response, several major technology companies and telecommunications service providers have become members of the OpenRAN Policy Coalition[2], whose mission is to "... promote policies that will advance the adoption of open and interoperable solutions in the Radio Access Network (RAN) as a means to create innovation, spur competition and expand the supply chain for advanced wireless technologies including 5G." Members come from many countries including the USA, India, Japan and Spain.
- **Policy Approaches:** Policy approaches to promoting sovereignty can placed roughly along the four dimensions: 1) isolationist, 2) cooperative, 3) competitive and 4) militaristic.

  The isolationist approach focuses on building and maintaining a nations digital infrastructure using domestically sourced components and the native workforce. This approach is largely limited to a small number of countries such as the US and China. A recent example of isolationist policy is that of Russia creating its own financial network (Mir) to avoid dependence on Visa and Mastercard [7] and pushing a homemade alternative to Youtube to citizens [8]. The Mir network is not broadly connected to the rest of the international financial clearing systems and the alternative to Youtube seems to have failed to gain user acceptance. An isolationist view may also cause technology stagnation (while the rest of the world moves to better technology).

  A cooperative policy approach seeks to form international agreements (e.g., treaties, open standards) to regulate the construction and operation of digital infrastructure. Policy efforts along these lines include the Clean Network initiative of the US under the Trump administration [9] which sought to form a group of countries building 5G cellular networks free of untrusted vendor hardware and software. In a similar vein, is the EU GAIA-X project to build a trusted networked cloud service based in the EU to support sovereign data services using open source software, certified hardware and verifiable exchanges. Recently (in July 2022), the USA struck an agreement with Saudi Arabia to invest in and deploy OpenRAN technologies for 5G cellular networks[3].

  The competitive approach aims to seek to form strategic partnerships between industry and government to promote digital infrastructure technology from domestic or trusted vendors. For example China's Belt and Road Initiative

[2]https://www.openranpolicy.org/

[3]https://www.politico.com/newsletters/weekly-cybersecurity/2022/07/18/will-the-saudis-help-the-u-s-beat-huawei-00046280

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provided grants and low cost loans to developing nations to build out digital infrastructure using Chinese vendors (e.g., Somalia data center).

Lastly militaristic methods revolve around deploying military resources to protect (physically and cyber) digital infrastructure and to serve as a retaliatory threat to adversaries.

- **Judicial Approaches:** The legal system in a nation or a group of nations can create laws and also adjudicate disputes, for example in SEP related issues in ways that support a nation's domestic interests [10].
- **Jurisdictional Issues:** International digital information flows are a particular point of contention in digital sovereignty discussions. The vast majority of international communications occurs over undersea cables. Furthermore, undersea cables enable domestic connectivity within some nation states (e.g., mainland US and Hawaiian islands, mainland Portugal and Azores and Madeira islands). However, the cables themselves are often in international waters and may be subject to physical or cyber attacks or unauthorized tapping without recourse to legal action [11]. As another example, Senegal used data centers in US and Asia via cables to store government data.

B. Resilience

Resilience has several contextual definitions in the scientific literature, a discussion of different definitions is given in [12]. In the communication network portion of digital infrastructure we note the International Telecommunications Union (ITU) defines *network resilience* [13] as "the ability to provide and maintain an acceptable level of service in the face of faults and challenges to normal operation of a given communication network, based on prepared facilities". In the broader context of a digital system providing a service, NIST has defined *cyber resiliency* [14] as "the ability to anticipate, withstand, recover from, and adapt to adverse conditions, stresses, attacks, or compromises on systems that use or are enabled by cyber resources". In this definition, a cyber resource is defined as an information resource which creates, stores, processes, manages, transmits, or disposes of information in electronic form and that can be accessed via a network or using networking methods.

Techniques for improving the resilience of digital infrastructure revolve around the concepts of segmentation and prioritization, adaptive response, redundancy, hardening, diversity and deception. Segmentation and prioritization categorizes infrastructure resources and services based on their trustworthiness and importance and supports mechanisms to separate various service resources if needed. Adaptive response involves reconfiguration of some of system resources in reaction to an attack or failure to maintain critical services. Redundancy is providing duplicate resources and services. Hardening encompasses techniques to make resources resistant to attack and failures, such as providing backup power supplies. Diversity refers to adopting heterogeneity in the infrastructure design and technology to reduce common vulnerabilities and other correlated issues such as supply chain and maintenance. Lastly, deception corresponds to hiding resources and promoting misleading information to adversaries.

A large number of metrics [15] have been proposed to quantify various aspect of digital infrastructure resilience ranging from measures of qualitative subjective user experience to quantitative measures of network connectivity. Consider the case of wide area terrestrial networks and under sea cable systems numerous connectivity metrics such as, the k connectivity, average number of routes, minimum maximum flow under any single link failure, have been developed and utilized [16]. Similarly, metrics for identifying the most import nodes and links of a network (e.g., node with highest betweenness centrality, node with highest degree centrality, etc.) have been proposed. The mapping between the existing resilience metrics and metrics for evaluating digital sovereignty needs to be investigated and clearly defined.

III. Sovereignty and Network Resilience

Implementing mechanisms to improve digital infrastructure resilience while ensuring sovereignty needs a framework that connects the two issues. In the absence of sovereignty concerns, a network can be made resilient in many ways that have been studied in the past [15]. With sovereignty concerns entering the picture, network resilience needs to be viewed differently. This is a toy example that only considers connectivity. The idea extends to issues such as equipment/node compromise (is the manufacturer, software support, etc. coming from a potential adversary), eavesdropping/confidentiality, and other cybersecurity concerns.

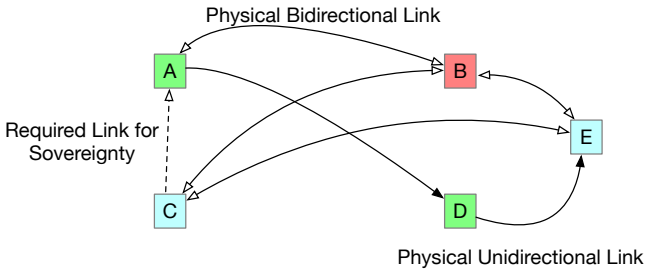


Fig. 1: Factors affecting Digital Sovereignty

a) Example 1 – Sovereignty Concerns: Consider Figure 1. In this figure, "friendly" nodes are colored green, "neutral" nodes are colored teal, and adversarial nodes are colored "red". Physical links are directional as shown in the figure (and assumed to have some redundancy e.g., power back-ups). Node A can communicate with Node E, both through adversarial node B and through a friendly node D, but node E has to rely on node B to get back to node A. Thus, if the relation between A and B were to deteriorate, there is a chance that E will be unable to communicate with A. One way of increasing sovereignty in this situation will be to create at least a unidirectional link between C and A.

b) Example 2 – Sovereignty and Resilience Interaction: Next we consider an example of the interaction between resilience and sovereignty. In networks that comprise critical infrastructure, it is common to have a *working or primary path*

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for "flows" between a source and destination node. If such a working path was to fail (due to accidents, component failures, disasters, cyber attacks), it is usually possible to allow part or all of the communication to continue through *disjoint* back-up or alternate paths. The term disjoint is important here, because this implies that the back-up paths are not affected at the same time as the working path.

In Figure 2 (top), we assume that every node in the network has acceptable "sovereignty" features (hence they are all gray). Let us suppose that node A has a working path $A-B-C-D$ for communication with node D. The back-up paths could be any one of these:

- Path 1: $A-E-F-C-D$
- Path 2: $A-G-H-D$
- Path 3: $A-G-E-F-H-D$
- Path 4: $A-E-F-H-D$
- Path 5: $A-G-H-F-C-D$
- Path 6: $A-E-G-H-D$.

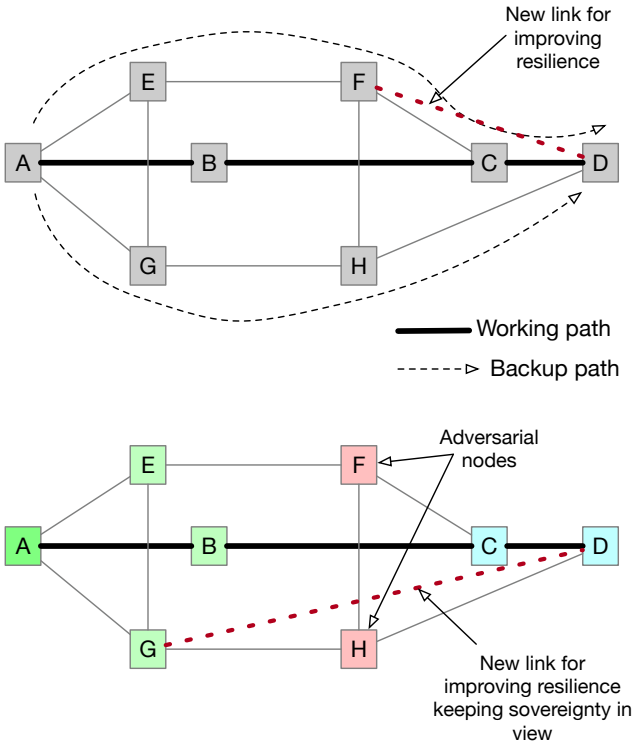


Fig. 2: Interaction between Resilience and Sovereignty: (top) agnostic of sovereignty concerns (bottom) with a consideration of sovereignty concerns

Note that traffic can continue to flow from node A to node D if nodes B or C fail or even if two links from node A (say to E and B) fail by selection of an unaffected backup path. In fact, the network topology is resilient to the loss of any single communication link or any single intermediate node in the set $\{B, C, E, F, G, H\}$. From a resilience-only point of view, it would be normal, if we are considering adding a single link, to add a link between nodes F and D to improve the connectivity of the network under failure.

Let us now suppose that nodes F and H belong to an adversarial nation (see Figure 2(bottom) where they are colored red). First, there are no disjoint back-up paths that avoid these two nodes. Thus, under failure of a node or link on the working path, node A will have to rely on one of these adversarial nodes to maintain communication with node D. Second, if the network design were to consider adding a new link, such a link may best be added between nodes E and D or G and D as shown in the figure in order to improved the resilience taking into consideration sovereignty concerns.

c) Example 3 – Cybersecurity Concerns: A third example we discuss here is the *innards* of a network. In Figures 1 and 2, we have only considered nodes in a generic network - usually a backbone spanning large distances. In Figure 3, let us consider node C, which is part of the working path from A to D. Although C is in a neutral territory, some components of the network in C may be coming from vendors in an adversarial country. Traffic flowing from A to D may be captured by such nodes upon a malicious software update (or even normal operations where data may be captured for performance reasons by the vendor). Under tense situations between the green and red nations, nodes may selectively drop packets or cause performance degradation[4]. Thus, the security of information may be compromised because of sovereignty - and the resilience of even the working path becomes suspect. Even when the working path is available, it may be beneficial to find alternative paths that do not involve nodes with components from adversaries for critical information.

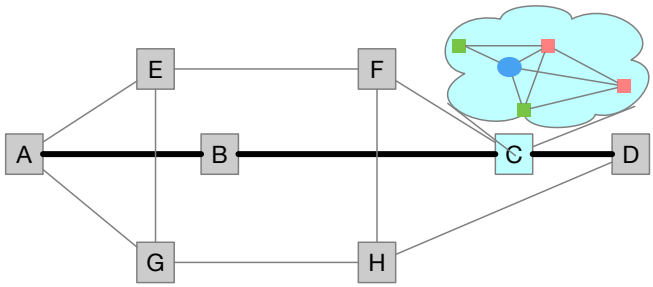


Fig. 3: Cyber-Dependencies - neutral node has adversarial components

d) Example 4 – Undersea Cables: Many major US based technology companies (e.g., Google, Microsoft) maintain software development centers in India, in addition the US is India's largest trading partner. The vast majority of communications between the US and India occurs via undersea cables which must take a multi-hop route either traversing the Pacific to Singapore then on to India or across the North Atlantic to Europe (France or UK) and then through the Red Sea to India. The current infrastructure provides a great deal of resilience with multiple cable routes currently in service and more planned with increase capacity [17]. However, from a sovereignty standpoint, the current set of cables fall within waters that China claims (i.e., South China Sea) [18] or pass through the politically risky Red Sea

[4]The situation may be considered similar to reduced natural gas flows to Europe from Russia during the recent war in Ukraine.

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region or are constructed with the aid of China or using Chinese vendors (e.g., HMN Tech). The lone exception to these "low-sovereignty" physical cables being the shared cable from the UK around the horn of Africa to India, this alternative however has limited capacity and long latency. Capacity and latency are important factors from networking standpoint and impact resilience significantly. Clearly from a sovereignty standpoint additional cables are need along geographically diverse routes.

IV. Open Problems

To design a more resilient network with a minimal level of sovereignty, several approaches may be considered. These include sovereignty enhancing methods such as building one's own network (e.g., Clean Network initiative), building it with coalition partners, creating international agreements, using verifiable or open standards, enhancing internal competition/research by subsidizing one's own companies to compete against adversarial vendors and so on. Militaristic approach to safeguard digital infrastructure and threat of force may be another option. In this section, we discuss some open problems and directions related to the interaction between digital sovereignty and resilience. These are by no means exhaustive, but suggest potential lines of enquiry in this problem space.

A. Digital Sovereignty Metrics

Currently there are not well established metrics and methods for quantifying digital sovereignty. As discussed in the previous sections and through the examples, there are several dimensions to the sovereignty problem. Quantifying the sovereignty metric is no simple task. We may consider indexes, such as the democracy index[5] or risk indexes such as, the Environmental, Social and Governance Index[6] associated with territories to assess potential threats to digital infrastructure components passing through, located in or manufactured or controlled by these territories.

However, suppose we are able to quantify digital sovereignty using a sovereignty factor s, (which may be probabilistic, but for now we assume it is a number that is monotonically increasing for fewer sovereignty concerns), then it may be possible to consider the design of networks and their components taking into account this factor. For example, consider one multi-hop route between nodes A and D in Figure 2 (top). Let us suppose that the probability of failure for each link is independently p and the sovereignty of each link is s. While the failure probabilities multiply in the computation of availability (all links must be available - hence no link must fail), the sovereignty may be $min_{s \in \text{path}}(s)$ since the most vulnerable link dominates the sovereignty concerns.

B. Trust and sovereignty

Countries in coalitions like NATO or Five Eyes may have higher trust amongst each other, thereby reducing concerns of losing sovereignty. However, recent work such as that in [19] indicate that other countries, not directly part of a coalition may be more reliable partners. It may be argued that a good *trust metric* may in fact be used in lieu of a sovereignty metric, since concerns about digital sovereignty often are based in the lack of trust and suspicion or geopolitical rivalry [20].

Subjective logic [21] which uses a tuple of trust, distrust and uncertainty may be a good approach to also develop a sovereignty metric. This logic has built in mechanisms for consensus (that combines concurring data), discounting, and propagation to update and aggregate trust values. Historical actions may be included in trust calculations as in [22] that considers IoT, although loss of digital sovereignty due to mistakes may take a long time to repair unlike trust in IoT.

C. Multi-layer resilience and sovereignty

In network design, often the resilience needs to be considered across the various layers of the stack. For example, a "link" that exists at the network (Internet Protocol - IP) layer typically depends on several links at the physical layer. Similarly, as we saw in Figure 3, although physically a node or link may be in sovereign territory or control, the software and vendor dependencies may make it vulnerable. How to incorporate physical layer sovereignty into higher layer connectivity is a problem worthy of study.

D. International Rules

As mentioned in Section III, agreements and treaties between individual nations and broader international regulations can act to address digital sovereignty concerns. However, the time frames for agreements to be approved at various levels as well as enforcing international regulations is challenging. Often, penalties and sanctions can be used as deterrents and incentives may be added to align the behaviour and effect behavior modifications to meet international obligations [23]. An understanding of the economics of this behavioral modifications in conjunction with digital infrastructure alternatives that a nation may rely upon if adversaries tamper with data flow or availability may facilitate better use of resources in the design of transnational digital infrastructure.

To the best of our knowledge, there are no good models to assist with this exercise. The use of Agent Based Models for simulating international relations [24] could be combined with sovereignty metrics, network and data center models and simulations to examine the long-term impact of international regulations and treaties on the resilience of communication networks in the context of national interests.

E. Other notions of sovereignty

Thus far, we have discussed the interaction between resilience and sovereignty for "nations" as we broadly imagine them to be – members of the United Nations, recognized by each other (except for territorial disputes) as sovereign entities over geographic areas and people within. There may be other notions of digital sovereignty, be it in states or provinces, or other groups such as organizational sovereignty. These may not have national security concerns, but may have local implications in terms of competition, intellectual property, etc.

[5]https://en.m.wikipedia.org/wiki/Democracy_Index
[6]https://risk-indexes.com

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In the United States, native American communities have a level of authority to govern themselves and the people within native American lands. Thus, in a lesser sense than national sovereignty, they do have some (local) sovereignty. Recently, there has been some studies on what information and communications technologies mean for native American populations and what property rights they may be able to exercise on networks and technologies in their geographical domains (see for example [25]). Resilience in this context may involve external partners (such as Internet Service Providers or cell phone providers) outside of native American lands.

Similarities and differences between nation-state sovereignty/resilience interactions and such interactions for smaller groups is a topic for further study.

V. Conclusions

In this paper the concept of digital sovereignty and how it interacts with digital infrastructure resilience was examined. A number of aspects to providing digital sovereignty in current digital infrastructure were examined, namely: physical dependencies and alliances, standards, vendor diversity, policy and judicial methods. Examples illustrating the interaction of resilience and sovereignty were given. Challenges to providing digital sovereignty while maintaining resilience were outlined including lack of metrics, reasoning about trust, the multi-layer nature of digital technology, international rules and regulations and of course cost constraints. Possible paths for future research to solve some of the identified challenges were examined and we note the intersection of digital sovereignty and resilient digital infrastructure is an area rich in open research problems with much work to be done.

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